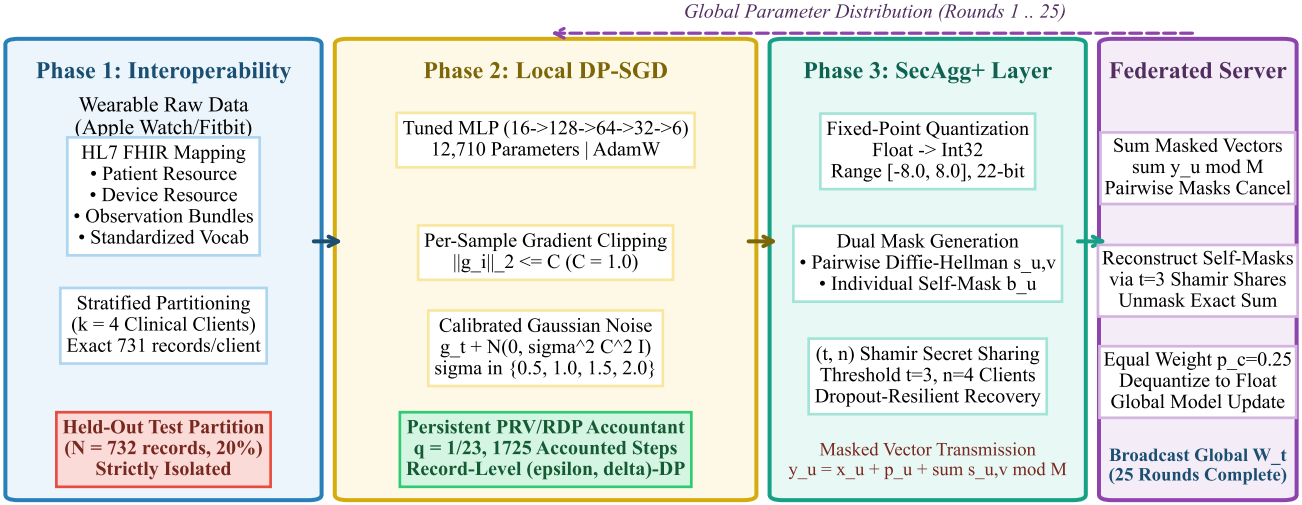

Global Parameter Distribution (Rounds 1 .. 25)
Phase 1: Interoperability
Wearable Raw Data
(Apple Watch/Fitbit)
HL7 FHIR Mapping
• Patient Resource
• Device Resource
• Observation Bundles
• Standardized Vocab
Stratified Partitioning
(k = 4 Clinical Clients)
Exact 731 records/client
Held-Out Test Partition
(N = 732 records, 20%)
Strictly Isolated
Phase 2: Local DP-SGD
Tuned MLP (16->128->64->32->6)
12,710 Parameters | AdamW
Per-Sample Gradient Clipping
||g_i||_2 <= C (C = 1.0)
Calibrated Gaussian Noise
g_t + N(0, sigma^2 C^2 I)
sigma in {0.5, 1.0, 1.5, 2.0}
Persistent PRV/RDP Accountant
q = 1/23, 1725 Accounted Steps
Record-Level (epsilon, delta)-DP
Phase 3: SecAgg+ Layer
Fixed-Point Quantization
Float -> Int32
Range [-8.0, 8.0], 22-bit
Dual Mask Generation
• Pairwise Diffie-Hellman s_u,v
• Individual Self-Mask b_u
(t, n) Shamir Secret Sharing
Threshold t=3, n=4 Clients
Dropout-Resilient Recovery
Masked Vector Transmission
y_u = x_u + p_u + sum s_u,v mod M
Federated Server
Sum Masked Vectors
sum y_u mod M
Pairwise Masks Cancel
Reconstruct Self-Masks
via t=3 Shamir Shares
Unmask Exact Sum
Equal Weight p_c=0.25
Dequantize to Float
Global Model Update
Broadcast Global W_t
(25 Rounds Complete)